

Important: Text generated by AI and may contain incorrect information, always check the content.

```
```python
Import necessary libraries
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

Check for missing values
display('Check for missing values:')
display(df.isnull().sum())

Check the structure of the DataFrame
display('DataFrame structure:')
display(df.head())
display(df.info())
display(df.describe())

Analyze categorical variables
categorical_cols = ['subreddit', 'post_id', 'sentence_range', 'text']
display('Summary statistics for categorical columns:')
for col in categorical_cols:
 display(f'Column: {col}')
 display(df[col].value_counts().head())

Analyze numerical variables
numerical_cols = ['label', 'confidence', 'social_timestamp']
display('Summary statistics for numerical columns:')
display(df[numerical_cols].describe())

Plot histograms for numerical variables
display('Histograms for numerical columns:')
for col in numerical_cols:
 plt.figure(figsize=(8,6))
 sns.histplot(df[col], kde=True)
 plt.title(f'Histogram of {col}')
 plt.show()

Plot bar charts for categorical variables
display('Bar charts for categorical columns:')
for col in categorical_cols:
 plt.figure(figsize=(8,6))
```



```
df[col].value_counts().head().plot(kind='bar')
plt.title(f'Bar chart of {col}')
plt.show()

Check correlation between numerical variables
display('Correlation between numerical columns:')
plt.figure(figsize=(8,6))
sns.heatmap(df[numerical_cols].corr(), annot=True, cmap='coolwarm')
plt.show()

Convert social_timestamp to datetime and extract year, month, day
df['social_timestamp_datetime'] = pd.to_datetime(df['social_timestamp'], unit='s')
df['year'] = df['social_timestamp_datetime'].dt.year
df['month'] = df['social_timestamp_datetime'].dt.month
df['day'] = df['social_timestamp_datetime'].dt.day

Analyze the distribution of label and confidence across subreddit
display('Distribution of label and confidence across subreddit:')
plt.figure(figsize=(8,6))
sns.boxplot(x='subreddit', y='label', data=df)
plt.title('Boxplot of label across subreddit')
plt.show()

plt.figure(figsize=(8,6))
sns.boxplot(x='subreddit', y='confidence', data=df)
plt.title('Boxplot of confidence across subreddit')
plt.show()
```
```

'Check for missing values:'

| | 0 |
|----------------|---|
| subreddit | 0 |
| post_id | 0 |
| sentence_range | 0 |
| text | 0 |
| label | 0 |





0

confidence 0

social_timestamp 0

dtype: int64
'DataFrame structure:'

| | subreddit | post_id | sentence_range | text | label | confidence | social_timestamp |
|---|------------------|---------|----------------|---|-------|------------|------------------|
| 0 | ptsd | 8601tu | (15, 20) | He said he had not felt that way before, sugge... | 1 | 0.8 | 1521614353 |
| 1 | assistance | 8lbrx9 | (0, 5) | Hey there r/assistance, Not sure if this is th... | 0 | 1.0 | 1527009817 |
| 2 | ptsd | 9ch1zh | (15, 20) | My mom then hit me with the newspaper and it s... | 1 | 0.8 | 1535935605 |
| 3 | relationships | 7rorpp | [5, 10] | until i met my new boyfriend, he is amazing, h... | 1 | 0.6 | 1516429555 |
| 4 | survivorsofabuse | 9p2gbc | [0, 5] | October is Domestic Violence Awareness Month a... | 1 | 0.8 | 1539809005 |



<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2838 entries, 0 to 2837
Data columns (total 7 columns):
Column Non-Null Count Dtype
--- ----- -----
0 subreddit 2838 non-null object
1 post_id 2838 non-null object
2 sentence_range 2838 non-null object
3 text 2838 non-null object
4 label 2838 non-null int64
5 confidence 2838 non-null float64
6 social_timestamp 2838 non-null int64
dtypes: float64(1), int64(2), object(4)
memory usage: 155.3+ KB
None

| | label | confidence | social_timestamp |
|--------------|-------------|-------------|------------------|
| count | 2838.000000 | 2838.000000 | 2.838000e+03 |
| mean | 0.524313 | 0.808972 | 1.518107e+09 |





| | label | confidence | social_timestamp |
|-----|----------|------------|------------------|
| std | 0.499497 | 0.177038 | 1.552209e+07 |
| min | 0.000000 | 0.428571 | 1.483274e+09 |
| 25% | 0.000000 | 0.600000 | 1.509698e+09 |
| 50% | 1.000000 | 0.800000 | 1.517066e+09 |
| 75% | 1.000000 | 1.000000 | 1.530898e+09 |
| max | 1.000000 | 1.000000 | 1.542592e+09 |

'Summary statistics for categorical columns:'
'Column: subreddit'

| | subreddit |
|------------------|-----------|
| ptsd | 584 |
| relationships | 552 |
| anxiety | 503 |
| domesticviolence | 316 |
| assistance | 289 |

dtype: int64
'Column: post_id'

| | post_id |
|--------|---------|
| 9md6qz | 6 |
| 7o5n2d | 6 |
| 7nnu34 | 5 |
| 7r6fsr | 5 |
| 6yq1rh | 5 |





dtype: int64
'Column: sentence_range'
sentence_range

| | |
|----------|-----|
| (0, 5) | 337 |
| [0, 5] | 334 |
| [5, 10] | 260 |
| (5, 10) | 215 |
| [10, 15] | 162 |

dtype: int64
'Column: text'

| | | text |
|--|--|------|
| <p>Hello, You are invited to complete a survey for a WMU psychology department research project designed to assess treatment preferences among adults seeking treatment for a variety of concerns. We hope to learn if and how preferences for treatment change over time, and if patient’s perception of their treatment’s match to their preferences is related to the benefit received from treatment. The survey is open to anyone ages 18 or older who is currently engaged in mental health treatment for at least one month and four therapy sessions, and not longer than one year. If you choose to participate, you will be asked to provide some demographic/background information, respond to survey items about your preferences for treatment and how they have changed over time, and about your quality of life. The survey may take between 30-40 minutes to complete.</p> | | 4 |
| <p>#NAME?</p> | | 4 |
| <p>Your responses to the survey items will be anonymous and kept confidential. Clicking the “SURVEY” link below will take you to a page asking you to read through a consent form explaining the purpose of this research, the content of the survey, the type of questions you will be asked, the amount of time it may take to complete the survey, and the risks and benefits of your participation. At the end of the form you can click “AGREE” to consent to the use of the answers you provide and to begin completing the survey. Thank you for your time and interest. <url></p> | | 3 |
| <p>7. This study will include only participants who are 18 years and older. Please send us a private message if you are interested in participating. Once you contact us, we will send you a link to the screening questionnaire to see if you qualify and to tell you more about the study. Please note: If you participated in this study at any point in 2016 through Vanderbilt University, unfortunately you are not eligible to participate again.</p> | | 3 |





If you have a survey you would like to share with us, you may do so here, please use the following format. Failure to do so will result in your survey being removed. Surveys not posted in here will result in a ban, the length of which will be decided at mods' pleasure. __Who I am__: _(Student, Researcher)_ __Affiliation__: _(university, company)_

2

dtype: int64

'Summary statistics for numerical columns:'

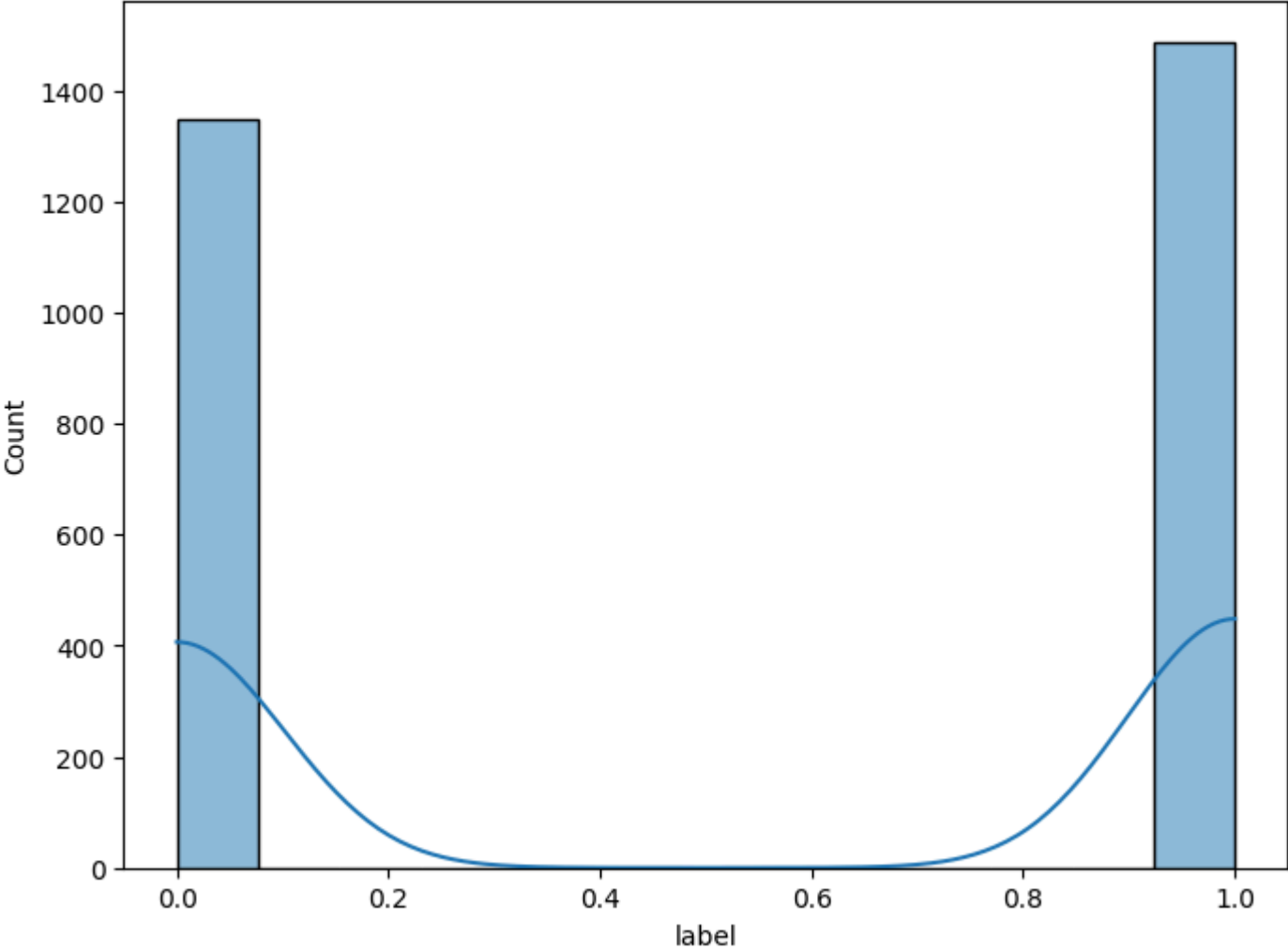
| | label | confidence | social_timestamp |  |
|-------|-------------|-------------|------------------|---|
| count | 2838.000000 | 2838.000000 | 2.838000e+03 | |
| mean | 0.524313 | 0.808972 | 1.518107e+09 | |
| std | 0.499497 | 0.177038 | 1.552209e+07 | |
| min | 0.000000 | 0.428571 | 1.483274e+09 | |
| 25% | 0.000000 | 0.600000 | 1.509698e+09 | |
| 50% | 1.000000 | 0.800000 | 1.517066e+09 | |
| 75% | 1.000000 | 1.000000 | 1.530898e+09 | |
| max | 1.000000 | 1.000000 | 1.542592e+09 | |

'Histograms for numerical columns:'



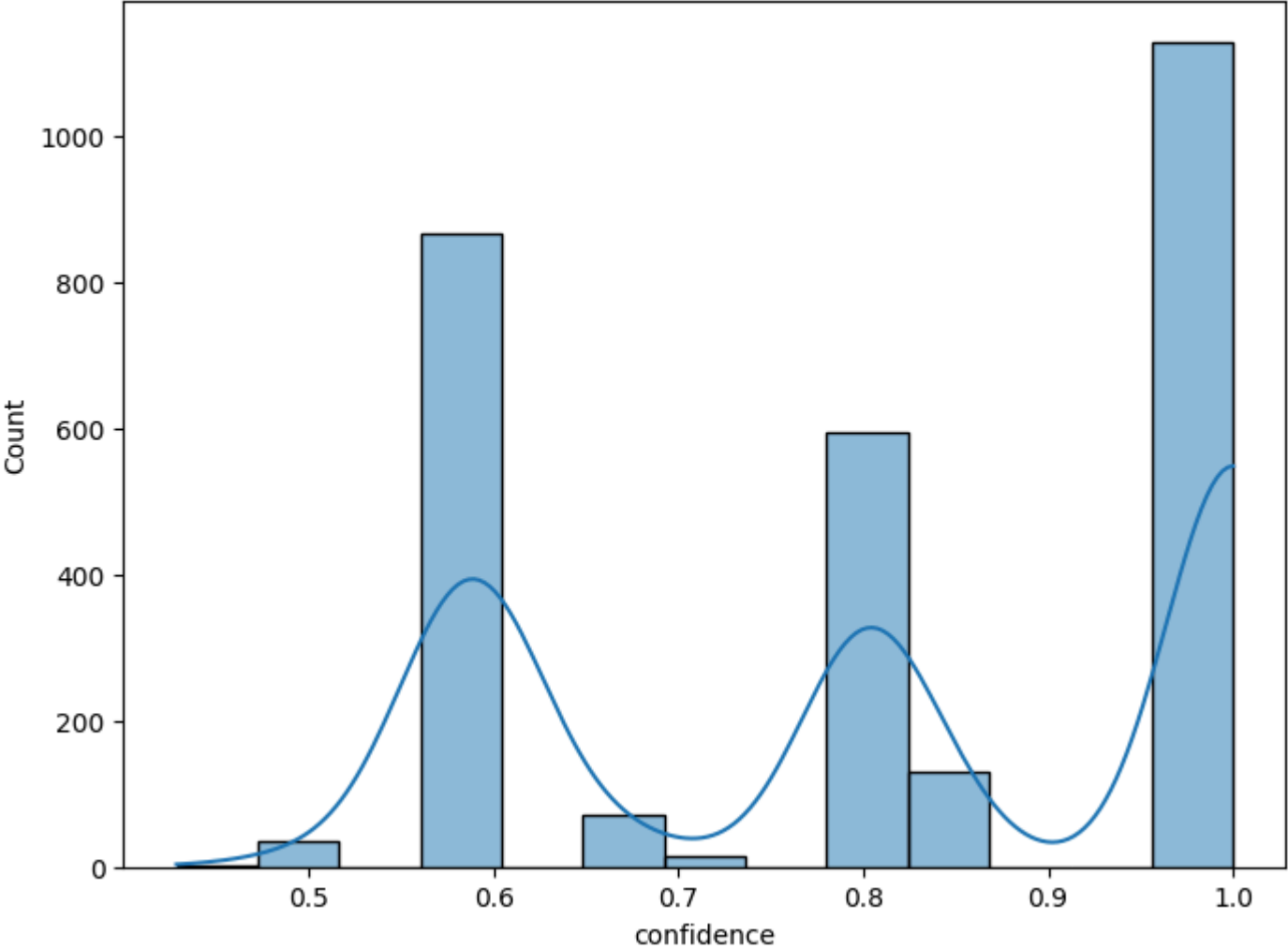


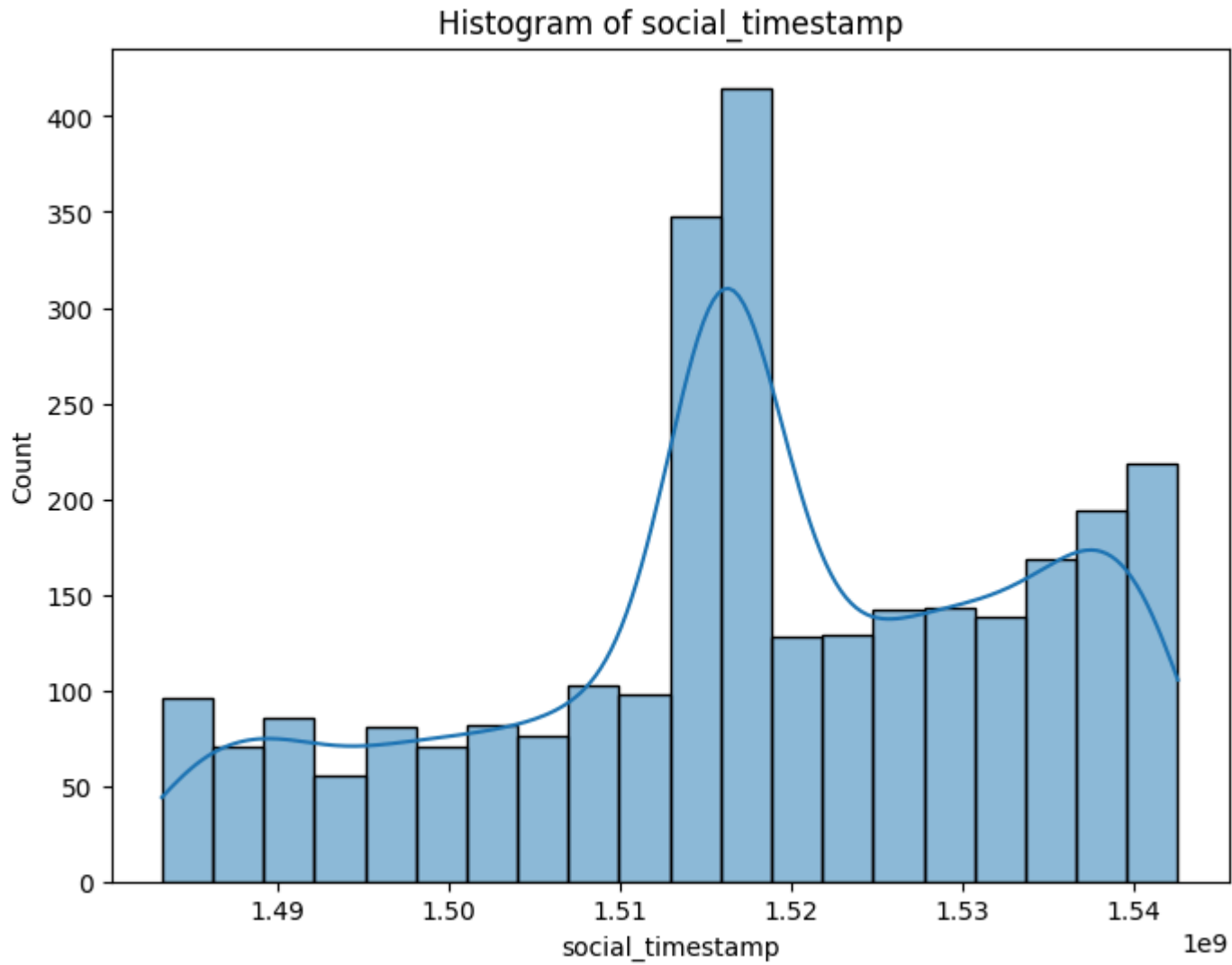
Histogram of label





Histogram of confidence

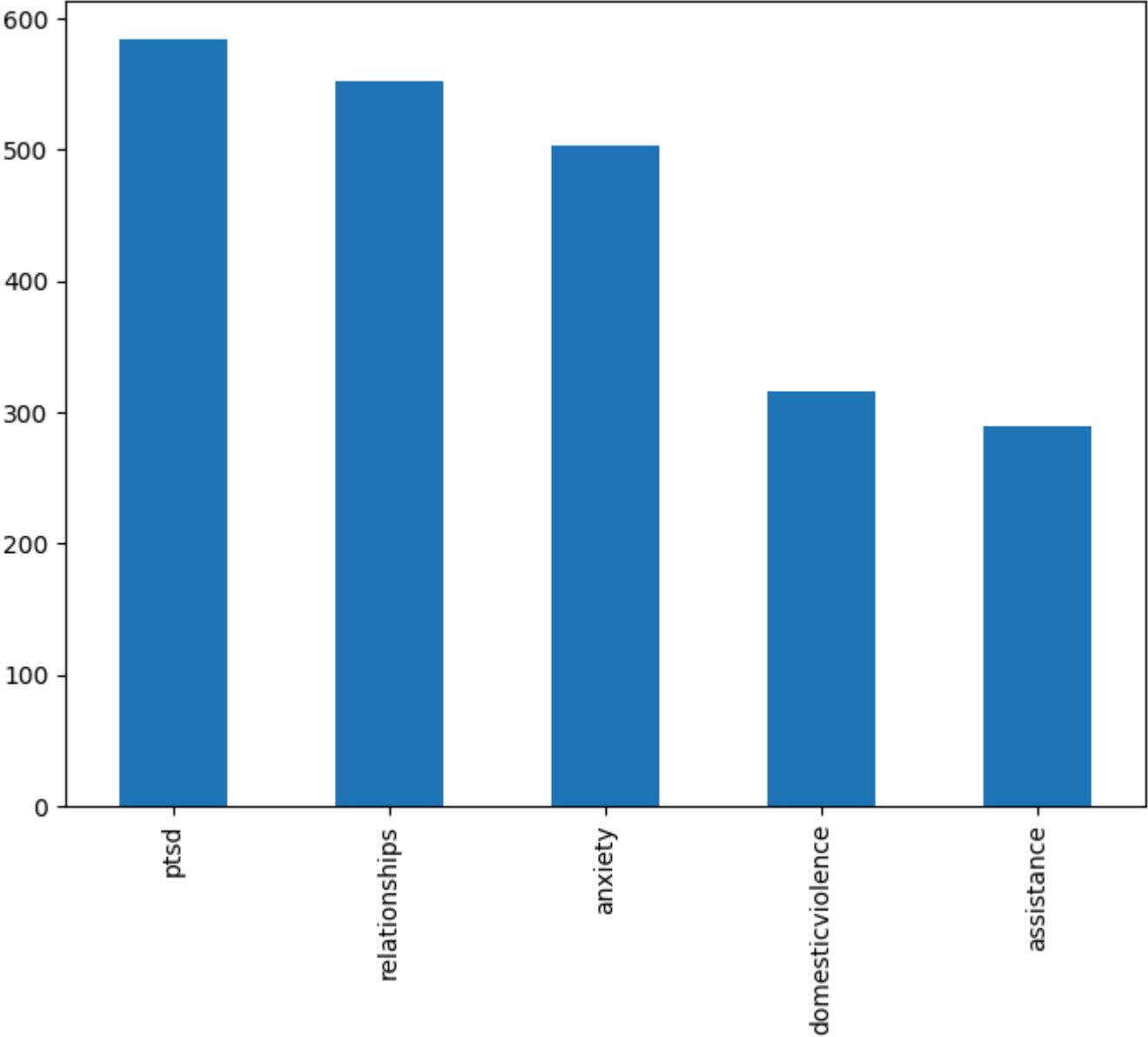




'Bar charts for categorical columns:'

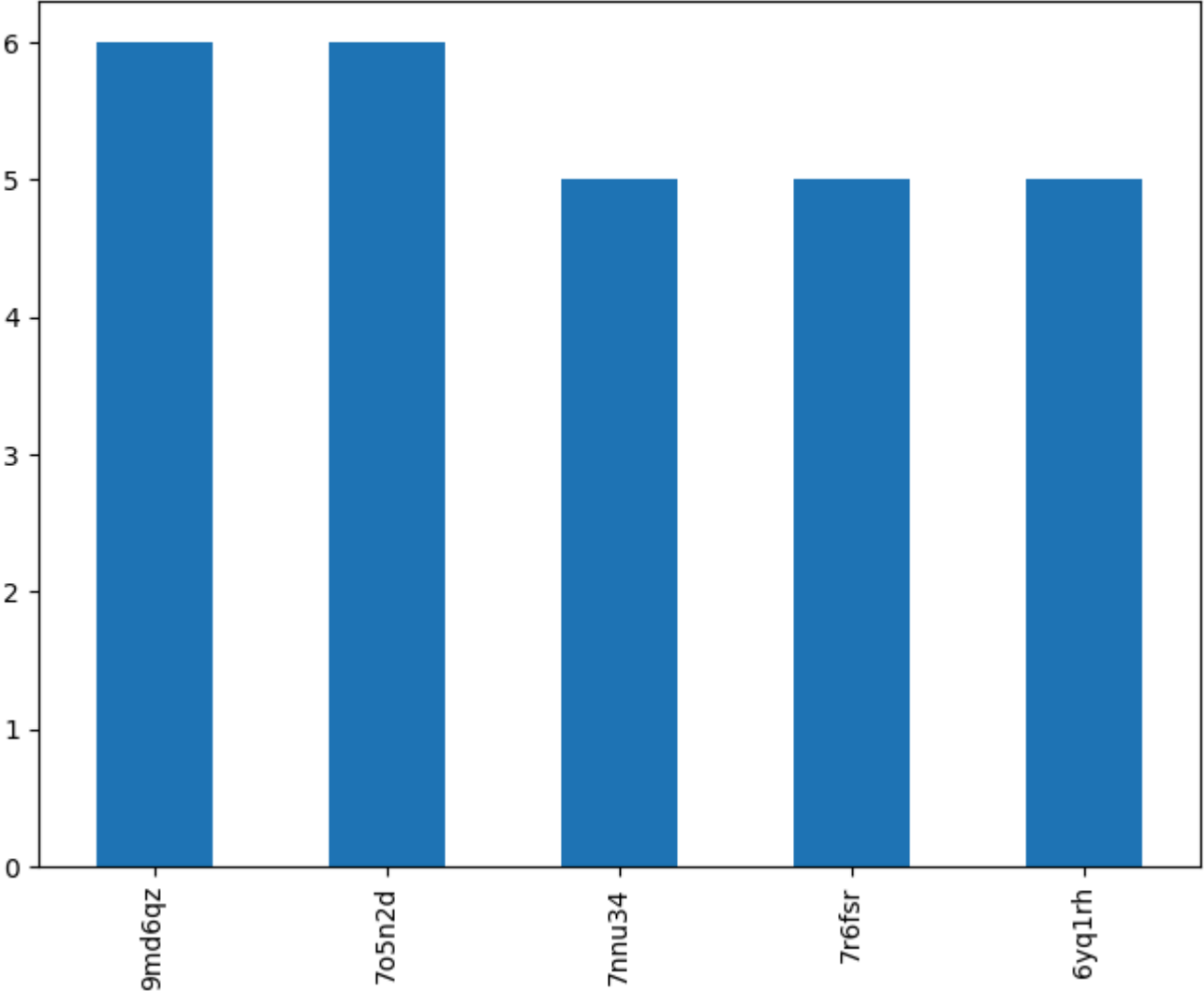


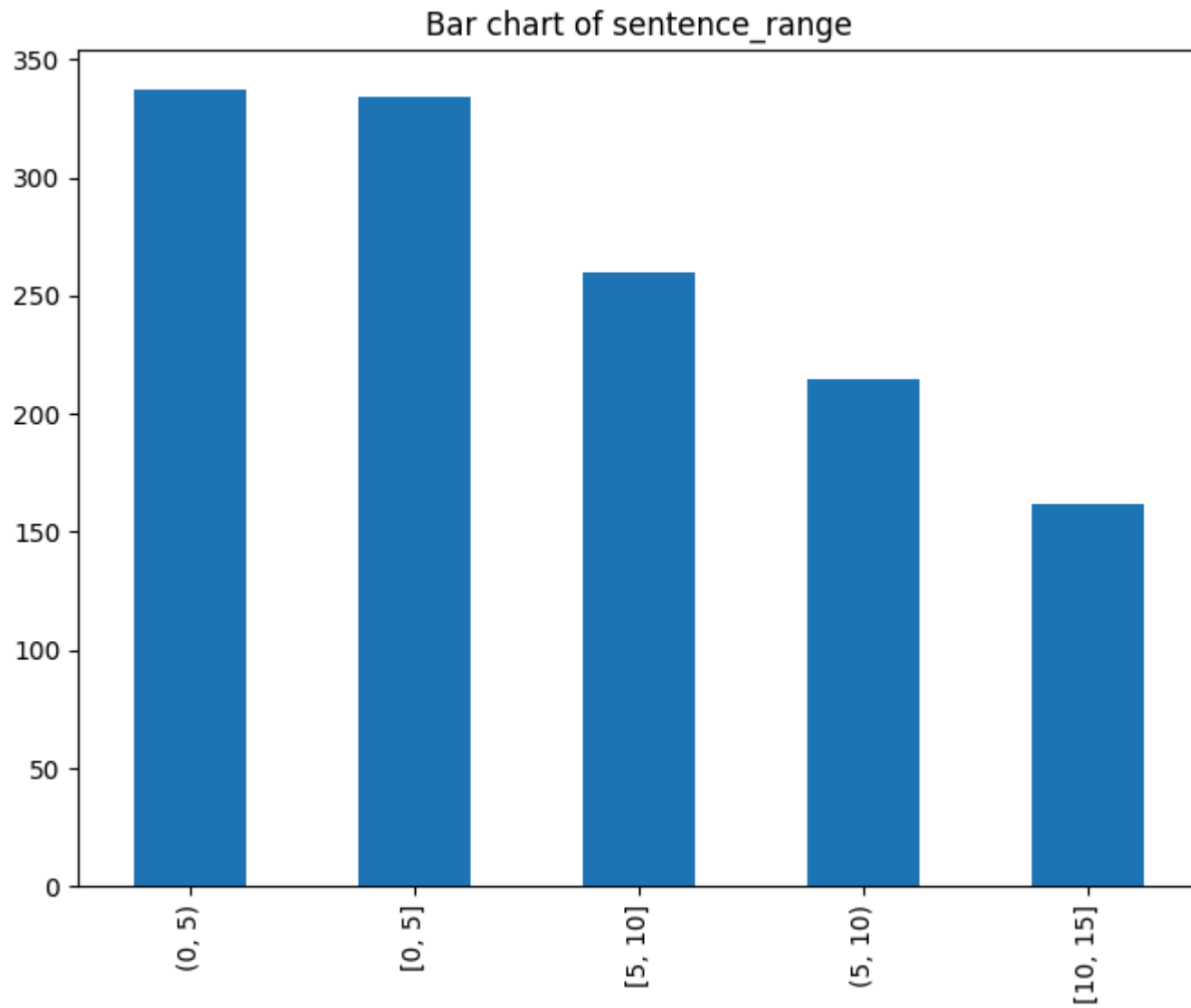
Bar chart of subreddit





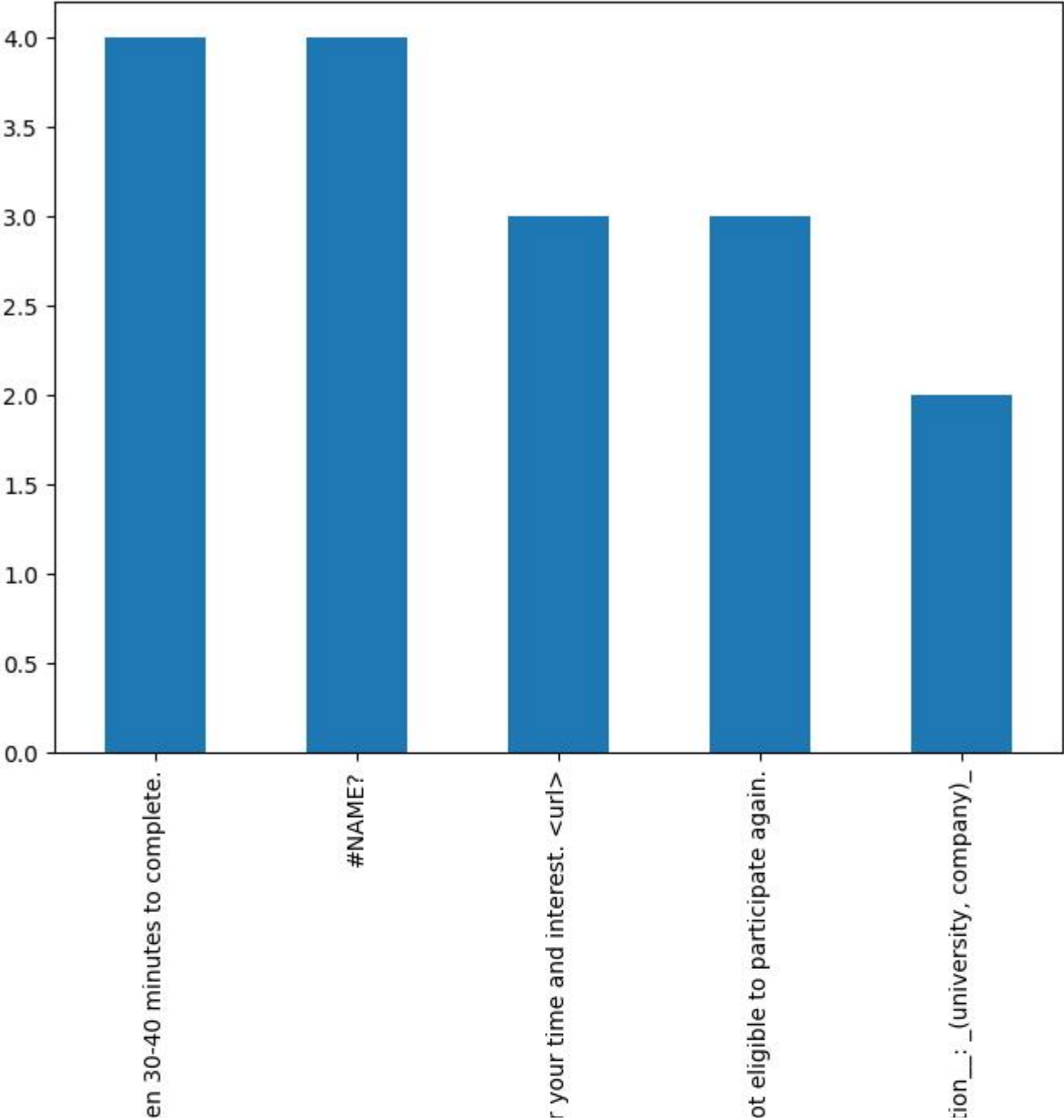
Bar chart of post_id







Bar chart of text



ment and how they have changed over time, and about your quality of life. The survey may take betwe

EE" to consent to the use of the answers you provide and to begin completing the survey. Thank you fo

ou participated in this study at any point in 2016 through Vanderbilt University, unfortunately you are n

, the length of which will be decided at mods' pleasure. __Who I am__:_(Student, Researcher)___Affilia

some demographic/background information, respond to survey items about your preferences for treati

the survey, and the risks and benefits of your participation. At the end of the form you can click "AGRI

the screening questionnaire to see if you qualify and to tell you more about the study. Please note: If yc

Failure to do so will result in your survey being removed. Surveys not posted in here will result in a ban

therapy sessions, and not longer than one year. If you choose to participate, you will be asked to provide

content of the survey, the type of questions you will be asked, the amount of time it may take to complete

a private message if you are interested in participating. Once you contact us, we will send you a link to

you have a survey you would like to share with us, you may do so here, please use the following format.

ges 18 or older who is currently engaged in mental health treatment for at least one month and four th

ou to a page asking you to read through a consent form explaining the purpose of this research, the coi

7. This study will include only participants who are 18 years and older. Please send us i

If y

h to their preferences is related to the benefit received from treatment. The survey is open to anyone a

the survey items will be anonymous and kept confidential. Clicking the "SURVEY" link below will take y

how preferences for treatment change over time, and if patient's perception of their treatment's matc

Your responses to

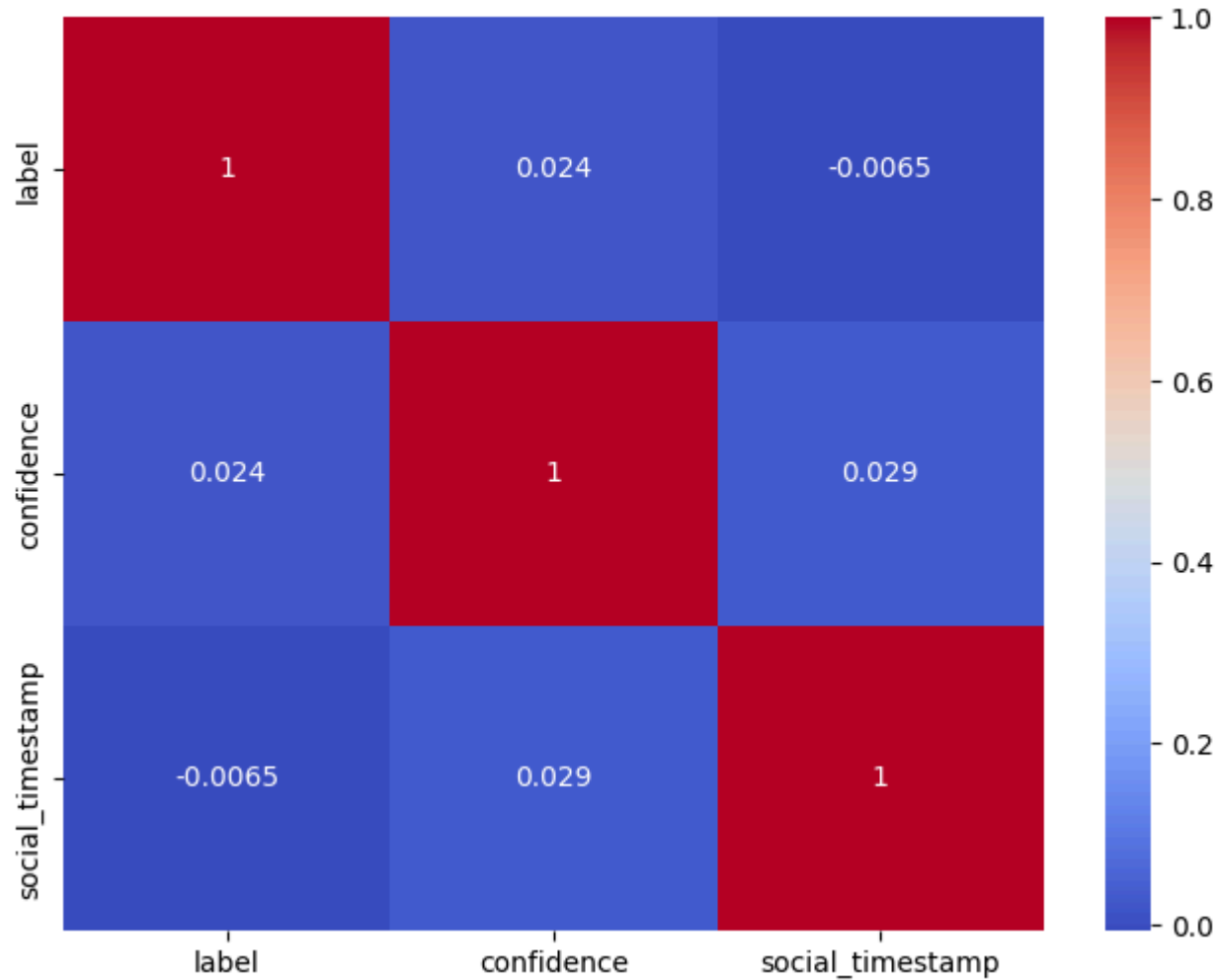
eatment preferences among adults seeking treatment for a variety of concerns. We hope to learn if and

invited to complete a survey for a WMU psychology department research project designed to assess tri



Hello, You are

'Correlation between numerical columns:'



'Distribution of label and confidence across subreddit:'



